

prbvolpath_sm — Sample Matching volume integrator in Mitsuba 3.9

Primal = same estimator as prbvolpath. All new machinery runs only in the adjoint pass.

Tests (gradient test: 48-seed adjoint $\langle g, \theta \rangle$ vs central FD, volpath referee. primal test: z-test vs volpath)

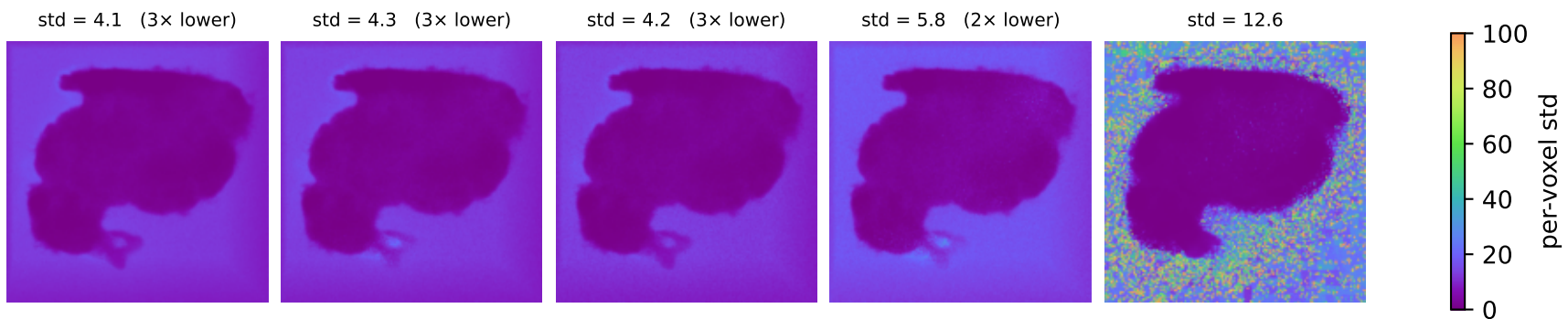
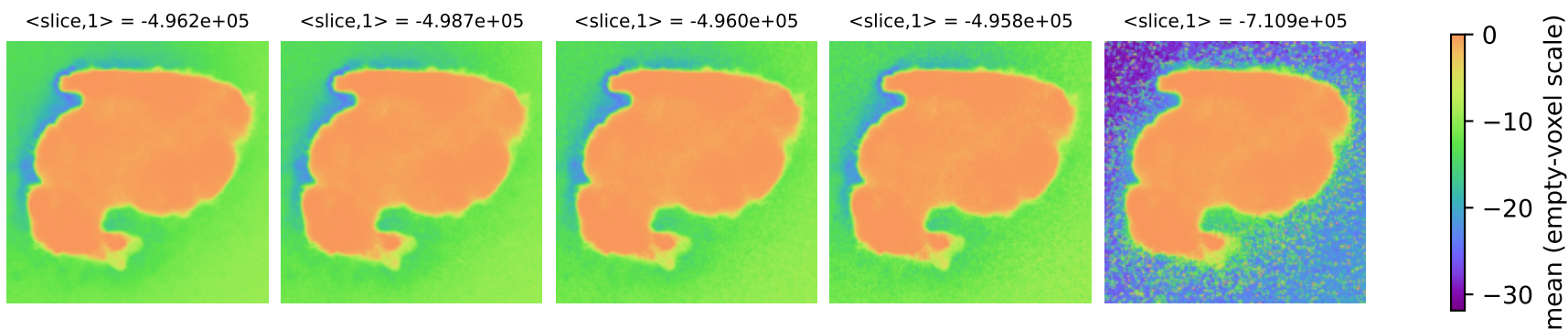
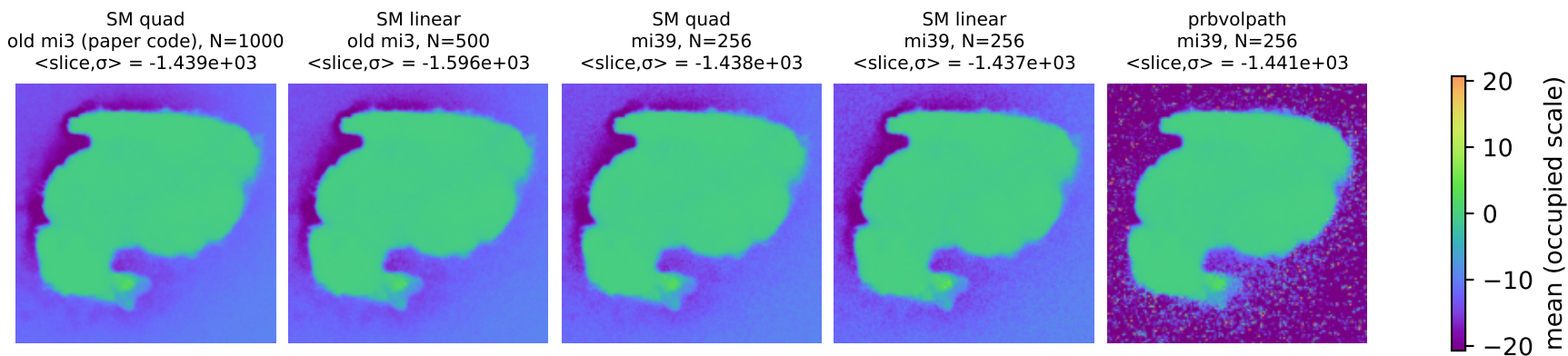
group	scenes	checked	# tests	result
boundary shapes	box / sphere / torus × quad / lin / K4	primal + σ grads	18	all pass
emitters	constant / envmap / area outside / area inside / point	primal + σ grads	30	all pass
media	homogeneous / no supergrid / albedo ramp / with surface	primal + σ + albedo	21	all pass
depth × RR	depth 16 / 64, RR on / off	σ grads	12	all pass
derivative sweep	3 shapes × 5 emitters (+ prbvolpath control column)	σ + albedo grads	120	all pass
phase function	HG $g = +0.5 / -0.3$	primal + g grads	14	all pass
light gradients	emitter radiance	grads	7	pass
empty voxels	carved zero-density voxels, $\langle g, 1 \rangle$ vs forward FD	σ grads, all-ones	4	3 pass, prb 2.3×*
LLVM backend	boundary + media + depth groups	mirror of CUDA	60	all pass
not supported	light/mesh position (needs edge sampling), camera pose	same as prbvolpath	—	documented

* *prbvolpath control: its detached estimator is intrinsically biased in $\langle g, 1 \rangle$ on dense-empty scenes (2.3× here, 1.7× bunny). Known, not an SM bug — SM is exact there.*

Bugs the stress tests caught (all fixed + committed)

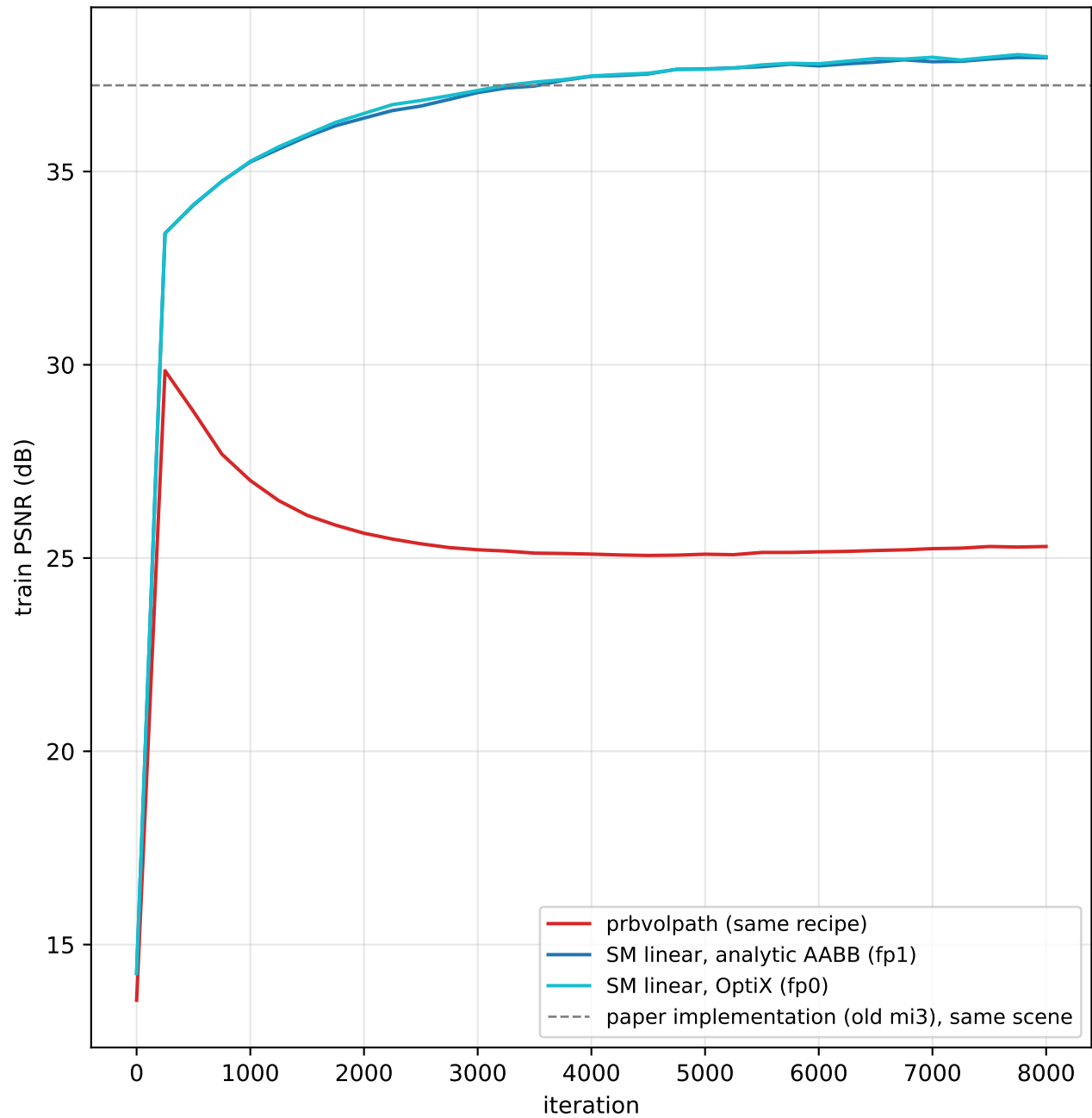
- probe domain vs grid bbox: Volume::eval clamps outside the grid, transport treats outside as vacuum. Probes now clipped to grid AABB. (was: +71% σ -grad bias on a torus boundary)
- dr.backward in evaluated code: default flags delete the persistent param→texture AD edge. Use ADFlag.ClearVertices. (was: albedo grads correct only for the first render per process)
- LLVM: eager backward on unevaluated loop-exit state drops gradients. dr.eval first. (was: −35% albedo, LLVM only)
- C++ walk left mei.wi / sh_frame = 0: anisotropic phase ran in a degenerate frame. (was: HG transport wrong, even primal — caught by the phase test)
- albedo as $\sigma_s / \max(\sigma_t, \epsilon)$: ratio = 0 in empty voxels, but the scatter derivative $d(\sigma_t \cdot \text{albedo}) / d\sigma_t = \text{albedo}$ is NOT. Use get_albedo(). (was: +69% $\langle g, 1 \rangle$ bias on the bunny; invisible to $\langle g, \sigma \rangle$ -weighted tests — new "goners" test: zero voxels + forward FD + $\langle g, 1 \rangle$)
- albedo-MIS density must be σ_s , not σ_t : the vertex estimator replays radiance that carries an albedo factor — at albedo = 0 it has no information, yet held weight $\sigma_t^2 / (1 + \sigma_t^2)$. (was: zero-albedo voxels lose that share of their albedo gradient, 0.969 → 0.999; prbvolpath: 0.904, same blind spot, no probe side to take over)

Same slice of $dI/d\sigma_t$ = backprop of sum(image), adjoint 1 per pixel. bunny cloud, uniform white light, 768×576, spp 8.
 $\langle \text{slice}, x \rangle = \Sigma \text{mean} \cdot x$ over the shown slice. Row 1: occupied-region scale, $x = \sigma$ (blind to empty voxels). Row 2: SAME mean, colorbar spanning the empty-voxel range, $x = 1$ — prbvolpath matches row 1 but is +43% off in row 2: its bias lives in the empty voxels.



256³ volume, occupied voxels, std: old quad 4.8 | old lin 5.2 | mi39 quad 5.0 | lin 6.9 | prb 22.5 → SM variance ÷20 (quad) / ÷10 (lin)

Inverse rendering: bunny cloud, L1, 8000 iterations, identical seeds + hyperparameters



final numbers

prbvolpath (same recipe)			
train	25.30 dB	test	24.63 dB 8.2 h
SM linear, analytic AABB (fp1)			
train	37.94 dB	test	35.56 dB 6.7 h
SM linear, OptiX (fp0)			
train	37.97 dB	test	35.67 dB 19.0 h
paper implementation (old mi3)			
train	37.23 dB	(3.6 h, reference framework)	

note: one shared lr schedule (the paper recipe) for all runs — not re-tuned per method. prbvolpath plateaus: high-variance σ gradients.

Inverse rendering, same held-out test view, spp 1024. PSNR = this view vs its own GT render.

GT (test view, held out)

old mi3 SM linear
paper code, 3.6 h
PSNR 37.23 dB

mi39 SM linear, analytic AABB
fp1, 6.7 h
PSNR 36.04 dB

mi39 SM linear, OptiX
fp0, 19 h
PSNR 36.17 dB

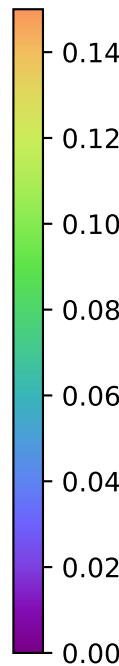
mi39 prbvolpath
same recipe, 8.3 h
PSNR 24.70 dB

|err|

|err|

|err|

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Speed (512², spp 8, seconds, A40. lower = better)

scene depth RR	prbvolpath	SM quad	SM lin	SM K4
blob bounce16 rroff	0.95 / 0.64	0.57 / 0.83	0.57 / 0.73	0.57 / 1.06
blob bounce64 rroff	0.96 / 0.66	0.58 / 0.86	0.58 / 0.75	0.58 / 1.09
blob bounce64 rron	0.89 / 0.59	0.53 / 0.74	0.53 / 0.68	0.53 / 0.98
dense bounce16 rroff	2.17 / 1.63	1.90 / 3.42	1.90 / 2.38	1.90 / 3.52
dense bounce64 rroff	3.55 / 2.94	3.05 / 8.01	3.05 / 4.13	3.05 / 6.02
dense bounce64 rron	1.61 / 1.17	1.41 / 2.38	1.41 / 1.74	1.42 / 2.61

table entry = primal / adjoint seconds. blob = 8³ sparse blob, peak σ 2, mean 0.5, mostly empty. dense = same scene, σ = 2.5 everywhere, deep scattering.

Notes

- SM primal is FASTER than prbvolpath everywhere: fused C++ free-flight walk (next page).
- SM adjoint: lin $\approx$ 1.1–1.5 $\times$ prbvolpath. quad / K4 pay more at high depth (more probes). RR helps a lot.
- bunny 768 $\times$ 576 spp16, per-kernel (OptiX): prb 3.86 s | quad 5.40 | K4 5.57 | lin 4.05. lin adjoint = replay 2.44 + probe flush 0.40 + one shared suffix 0.85.

Design notes (what is different from prbvolpath)

Estimator (the paper)

- σ_t has two opposite gradient terms: more scatter (+) and less transmittance (-).
- Sample matching: evaluate BOTH at the SAME uniform probe points on each path segment.
- Shared points → negative correlation → much lower variance. Unbiased.
- Variants: quad = one radiance suffix per segment ($O(n^2)$). lin = one suffix per path, reservoir-picked ($O(n)$). K4 = 4-slot striped reservoir, exact for paths ≤ 4 segments.

Deferred probes (two-stage adjoint)

- Problem: probes trace rays. Rays inside the big replay kernel → OptiX saves 128 registers + 3 KB/thread around every optixTrace. Very slow.
- Fix: replay loop only APPENDS a 68-byte record per segment (dr.scatter_inc, compacted).
- After the loop: read the counter, gather records, run all probes in one small separate kernel.
- Result: SM linear adjoint 26.5 s → 4.1 s on the bunny.

Fused C++ free-flight walk (Medium::sample_real_interaction_fused)

- One C++ loop fuses the majorant-supergrid DDA and accept-until-real null tracking.
- The Python loop body only ever sees a REAL scatter or an escape.
- Null collisions never spin the fat kernel → warp stays aligned on the expensive body.
- This is also why SM primal beats prbvolpath primal.
- Careful: fill the WHOLE MediumInteraction (wi, sh_frame, wavelengths, time, majorant...).

Reservoir without color bias

- DRT-style reservoir picks with mean-of-ratios → small color bias in colored media.
- Ours: scalar selection weight $v = \text{mean}(w)$, additive → pick probability exactly v/V .
- Compensate per channel: $w \cdot V / v$. Bias gone (measured 35% → 0.02%).

Dr.Jit interaction of the deferred (evaluated-mode) adjoint

- The deferred flush calls dr.backward outside any symbolic loop → uses flags=ADFlag.ClearVertices to keep the persistent param→texture edges alive across renders.
- Loop outputs feeding the linear suffix are dr.eval-ed before its backward (required on the LLVM backend).